

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024****Course: B.Tech Branch : Mechanical Engineering / Mechanical Engineering(Sandwich)**
/Mechatronics**Semester :VII Subject Code & Name: Name: BTMOE704C Plant Maintenance****Max Marks: 60****Date: 12/02/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Preventive maintenance consists of the following action.	CO1	1
	a. Replacement of components b. Repair of components. c. Service of components d. All of the above.		
2	Which is known as corrective maintenance?	CO1	1
	a. Breakdown maintenance b. Scheduled maintenance. c. Predictive maintenance. d. Preventive maintenance		
3	Which of the following is an advantage of plant maintenance?	CO4	1
	a. It decreases plant safety b. It increases life of machinery c. It decreases accuracy of machinery d. It decreases life of machinery		
4	What type of maintenance is most effective?	CO3	1
	a. Shutdown maintenance b. online maintenance c. Corrective maintenance d. Preventing maintenance		
5	In reciprocating compressor pressure pressure not increases due to---	CO5	1
	a. Suction valves working properly b. Suction line filter choke up c. Suction filter in good condition. d. Delivery valves working properly		
6	Shut down maintenance Consist of the following actions.	CO4	1
	a. Service of component b. Replace if component c. Repair of component d. All of the above		
7	----- Maintenance requires monitoring plant equipment health.	CO4	1
	a. Predictive b. online c. breakdown. d. Corrective		
8	What are the various possible causes for a failure of mechanical equipment?	CO5	1

	a. Manufacturing defects	b. Workmanship	c. incorrect usage of equipment	d. All of the above		
9	Due to lake of maintenance, Inventory cost_____				CO4	1
	a. Becomes low	b. Becomes high	c. Remains same	d. All of the above		
10	The Cost Which is done to keep the Plant, Machines & Equipments in proper Working Conditions is known as-----.				CO4	1
	a. Depreciation	b. Maintenance cost	c. Labour cost	d. Material cost		
11	----- techniques is not used in plant condition monitoring				CO6	1
	a. Vibration analysis	b. Casting	c. Ultra sound technology	d. Electrical testing.		
12	-----Is not elements of plant condition monitoring?				CO6	1
	A. Data collection	b. Break down	c. Continuous improvement.	d. Maintenance planning		
Q. 2	Solve the following.					12
A)	Explain briefly different types of plant maintenance?				CO1	6
B)	Explain in detail necessity of shut down maintenance				CO1	6
Q.3	Solve the following.					12
A)	Explain in detail preventive maintenance and its importance.				CO3	6
B)	Explain briefly Online maintenance.				CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Explain repair cycle of plant maintenance.				CO2	6
B)	Explain briefly maintenance of boiler.				CO5	6
C)	Explain Maintenance fault finding and troubleshooting techniques.				CO5	6
Q.5	Solve Any Two of the following.					12
A)	What is importance of stand by unit?				CO3	6
B)	What is mean of trouble shooting?				CO4	6
C)	Write colour code with its specifications				CO2	6
Q. 6	Solve Any Two of the following.					12

A)	Explain different types of technique for condition monitoring	CO6	6
B)	What are the benefits of condition monitoring?	CO4	6
C)	Explain maintenance of pump and compressor briefly.	CO5	6
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